ASSIGNMENT – 8 kd2_vaibhav_96218

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
kd2_vaibhav_96218>SELECT * FROM ORDERS
    -> WHERE cnum IN (SELECT CNUM FROM CUSTOMERS
    ->                 WHERE  CNAME = 'CISNEROS');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.066 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
kd2_vaibhav_96218>SELECT o.amt, c.cname, c.rating FROM customers c
    -> INNER JOIN orders o ON c.cnum = o.cnum
    -> WHERE o.amt > (SELECT avg(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.007 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
kd2_vaibhav_96218>SELECT snum, sum(amt) FROM orders
    -> GROUP BY snum
    -> HAVING sum(amt)>(SELECT max(amt) FROM orders);
+------+----------+
| snum | sum(amt) |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.011 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
kd2_vaibhav_96218>SELECT * FROM CUSTOMERS
    -> WHERE RATING >= ANY(SELECT RATING
    ->                          FROM CUSTOMERS
    ->                          WHERE SNUM = ( SELECT SNUM
    ->                                          FROM SALESPEOPLE
    ->                                          WHERE SNAME ='SERRES'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.014 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL

```
kd2_vaibhav_96218>SELECT snum, sname, city FROM SALESPEOPLE s
    -> WHERE S.CITY <> ALL( SELECT CITY FROM CUSTOMERS c WHERE c.snum = s.snum );
+------+--------+----------+
| snum | sname  | city     |
+------+--------+----------+
| 1004 | Motika | London   |
| 1007 | Rifkin | Barcelona |
| 1003 | Axelrod | New York |
+------+--------+----------+
3 rows in set (0.011 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
kd2_vaibhav_96218>select * from orders
    -> where amt > any (select amt from orders
    ->                  where cnum IN (select cnum from customers
    ->                                 where city = 'london'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.008 sec)
```

7. Extract all the orders of Motika.

```
kd2_vaibhav_96218>SELECT * FROM ORDERS
    -> WHERE SNUM = (SELECT SNUM FROM SALESPEOPLE
    ->                  WHERE SNAME = 'MOTIKA');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.008 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
kd2_vaibhav_96218>SELECT * FROM orders
    -> WHERE SNUM IN (SELECT SNUM FROM CUSTOMERS
    ->                  WHERE CITY = 'LONDON');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.013 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
kd2_vaibhav_96218>SELECT S.SNAME, S.SNUM FROM SALESPEOPLE s INNER JOIN  customers c
 ON s.snum = c.snum
    -> group by s.sname, s.snum
    -> having count(c.cnum)>1;
+--------+------+
| SNAME  | SNUM |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.016 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
kd2_vaibhav_96218>SELECT s.snum, s.sname, s.city, count(c.cnum)
    -> FROM salespeople  s INNER JOIN customers c ON c.snum = s.snum
    -> GROUP BY s.snum, s.sname, s.city
    -> HAVING count(c.cnum) >=2;
+------+--------+----------+---------------+
| snum | sname  | city     | count(c.cnum) |
+------+--------+----------+---------------+
| 1001 | Peel   | London   |             2 |
| 1002 | Serres | San Jose |             2 |
+------+--------+----------+---------------+
2 rows in set (0.008 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
kd2_vaibhav_96218>SELECT * FROM customerS
    -> WHERE rating > ANY(SELECT min(rating) FROM customers
    ->                          WHERE CITY = 'ROME');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.005 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
kd2_vaibhav_96218>SELECT * FROM orders
    -> WHERE amt >ANY(SELECT amt FROM orders
    ->                 WHERE ODATE = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.038 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose

```
kd2_vaibhav_96218>SELECT * FROM orders
    -> WHERE amt < ANY (SELECT amt FROM orders
    ->                  WHERE cnum IN (SELECT cnum FROM customers
    ->                                  WHERE city = 'san jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.018 sec)
```

14. Select those customers whose rating are higher than every customer in London

```
kd2_vaibhav_96218>SELECT * FROM customers
    -> WHERE rating >ALL (SELECT rating FROM customers
    ->                    WHERE city ='LONDON');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.008 sec)
```